Business Analytics Case Study Report

Telco Customer Churn Analysis

AnalystLab Africa Data Analytics Internship — Batch C — Week 5

Introduction

Customer churn — when a customer stops doing business with a company — is one of the most significant challenges facing subscription-based businesses such as telecommunications providers. Acquiring a new customer typically costs far more than retaining an existing one, which makes understanding why customers leave a critical business priority. This report applies data analytics techniques to the Telco Customer Churn dataset to uncover the key factors driving customer attrition and to recommend practical, data-driven actions the business can take to reduce churn.

Business Problem Statement

The business challenge addressed in this case study is customer churn at a telecommunications company. Specifically, the analysis seeks to answer:

- Why are customers leaving the company?
- Which factors contribute most to customer churn?

Understanding these drivers allows the business to move from reactive customer-loss tracking to proactive retention strategy, targeting the customer segments most at risk before they churn.

Dataset Description

- Source: WA_Fn-UseC_-Telco-Customer-Churn.csv (Telco Customer Churn dataset, Kaggle)
- Rows: 7,043 customers
- Key fields analyzed: Contract type, Churn status
- Other available fields: tenure, monthly charges, internet service, payment method, and additional customer account details

Data Analysis Process

The analysis was carried out in Microsoft Excel using the following steps:

- Reviewed the dataset structure, confirming 7,043 rows and identifying the Contract and Churn fields as the primary variables of interest
- Built a pivot table summarizing the count of churned customers grouped by contract type (Month-to-month, One year, Two year)
- Calculated the churn rate for each contract type as: (customers who churned) ÷ (total customers in that contract type)
- Created a bar chart, "Churn Rate by Contract Type," to visualize and compare the three groups

Key Findings

The churn rate varies dramatically by contract type:

Contract Type	Customers	Churn Rate
Month-to-month	3,875	~43%
One year	1,473	~11%

Two year	1,695	~3%

Month-to-month customers churn at roughly 14 times the rate of two-year contract customers. This is by far the strongest pattern in the dataset and points to contract commitment as a major lever influencing customer retention. Customers with no long-term commitment appear far more willing to leave, likely because they face no penalty or switching cost for doing so.

Business Insights

Beyond the raw churn numbers, this analysis points to a few broader business insights:

- Contract length functions as a proxy for customer commitment — customers who commit to longer terms are far less likely to leave, suggesting that switching costs and psychological commitment both play a role in retention.
- Month-to-month customers represent the highest-risk segment and the largest group in the dataset, meaning even small improvements in this group's retention could have an outsized impact on overall churn.
- Because churn is concentrated so heavily in one segment, retention efforts can be tightly targeted rather than applied broadly across the entire customer base, making interventions more cost-effective.
- The sharp drop-off between month-to-month and one-year contracts suggests that even a modest step up in commitment (not necessarily a full two-year term) could meaningfully reduce churn.

Recommendations

Based on these findings, the following actions are recommended:

- Introduce loyalty programs that reward customers for staying on longer-term contracts.
- Offer incentives (discounts, added services) to encourage month-to-month customers to upgrade to one- or two-year contracts.
- Improve customer support services, particularly for month-to-month customers, who may be more price- or service-sensitive.
- Proactively flag and target month-to-month customers with retention offers before their renewal decision point.
- Monitor churn rate by contract type on an ongoing basis to track the impact of retention initiatives over time.

These recommendations are practical, directly supported by the analysis, and focused on the single strongest churn driver identified in the dataset: contract type.

Conclusion

This case study demonstrates that a focused, well-visualized analysis of a single key variable — contract type — can reveal a clear and actionable business insight. By encouraging customers toward longer-term contracts through loyalty incentives and improved support, the business has a concrete, data-backed opportunity to significantly reduce customer churn.